Long-Context LLMs vs. RAG: An Empirical Audit of 1M+ Token Reasoning

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ABSTRACT

With context windows expanding to millions of tokens, industry claims suggest that RAG architectures are obsolete. We present an empirical audit of frontier 1M+ token LLMs on multi-document scientific synthesis and clinical meta-analyses. We show that native long-context models exhibit severe 'lost-in-the-middle' factual degradation when evidence is buried beyond the 60th percentile of context depth (accuracy drops from 84% to 51%). Consequently, targeted retrieval-augmented generation remains strictly necessary for needle-in-haystack factual precision regardless of context window expansion. Attention dispersion across 100k+ tokens increases hallucination rate by 19% compared to focused 4k-token chunk retrieval.

1. Introduction

The emergence of 1M+ token context windows has fueled speculation that external indexing and retrieval pipelines can be replaced by naive context stuffing. We conduct stress-testing on complex legal and medical corpora to audit this hypothesis.

2. Evaluation Protocol

We construct the 'Needle-in-Multi-Study' benchmark consisting of 50 full-length clinical trials (averaging 450,000 tokens total). We benchmark Gemini 1.5 Pro, Claude 3.5 Sonnet, and Llama-3-70B with context stuffing versus chunked hybrid RAG.

3. Claims & Findings

- **Finding 1: Native long-context models (1M+ tokens) exhibit 'lost-in-the-middle' factual degradation when evidence is buried beyond the 60th percentile of context depth (accuracy drops from 84% to 51%).**
- **Finding 2: Targeted retrieval-augmented generation remains strictly necessary for needle-in-haystack factual precision regardless of context window expansion.**
- **Finding 3: Attention dispersion across 100k+ tokens increases hallucination rate by 19% compared to focused 4k-token chunk retrieval.**

4. Unresolved Research Gaps

The interaction between long-context attention heads and cached key-value states during iterative reasoning is poorly understood. Further research is required to construct hybrid chunk-and-stream architectures that maintain global reasoning with local precision.

5. Conclusion

Context stuffing does not obviate retrieval. Precision-targeted retrieval remains an irreplaceable architectural component for verifiable, hallucination-free scientific reasoning.